

CHAPTER 28

WIND LOADS ON BUILDINGS: MAIN WIND FORCE RESISTING SYSTEM (ENVELOPE PROCEDURE)

28.1 SCOPE

28.1.1 Building Types. This chapter applies to the determination of main wind force resisting system (MWFRS) wind loads on low-rise buildings using the Envelope Procedure.

Part 1 applies to all low-rise buildings where it is necessary to separate applied wind loads onto the windward, leeward, and sidewalls of the building to properly assess the internal forces in the MWFRS members.

Part 2 applies to a special class of low-rise buildings designated as enclosed simple diaphragm buildings as defined in Section 26.2.

28.1.2 Conditions. The design wind loads determined in accordance with this section shall apply to buildings complying with all of the following conditions:

1. The building is a regular-shaped building as defined in Section 26.2.
2. The building does not have response characteristics that make it subject to across-wind loading, vortex shedding, instability caused by galloping or flutter, nor does it have a site location for which channeling effects or buffeting in the wake of upwind obstructions warrant special consideration.

28.1.3 Limitations. The provisions of this chapter take into consideration the load magnification effect caused by gusts in resonance with along-wind vibrations of flexible buildings. Buildings that do not meet the requirements of Section 28.1.2 or that have unusual shapes or response characteristics shall be designed using recognized literature documenting such wind load effects or shall use the Wind Tunnel Procedure specified in Chapter 31.

28.1.4 Shielding. There shall be no reductions in velocity pressure caused by apparent shielding afforded by buildings and other structures or terrain features.

PART 1: ENCLOSED AND PARTIALLY ENCLOSED LOW-RISE BUILDINGS

User Note: Use Part 1 of Chapter 28 to determine the wind pressure on the MWFRS of enclosed, partially enclosed, or open *low-rise buildings* that have a flat, gable, or hip roof. These provisions use the Envelope Procedure by calculating wind pressures from the *specific equation* applicable to each building surface. For building shapes and heights for which these provisions are applicable, this method generally yields the lowest wind pressure of all of the analytical methods specified in this standard.

28.2 GENERAL REQUIREMENTS

The steps required for the determination of MWFRS wind loads on low-rise buildings are shown in Table 28.2-1.

**Table 28.2-1 Steps to Determine Wind Loads on MWFRS
Low-Rise Buildings**

-
- Step 1:** Determine risk category of building; see Table 1.5-1.
- Step 2:** Determine the basic wind speed, V , for applicable risk category; see Figs. 26.5-1 and 26.5-2.
- Step 3:** Determine wind load parameters:
- Wind directionality factor, K_d ; see Section 26.6 and Table 26.6-1.
 - Exposure category B, C, or D; see Section 26.7.
 - Topographic factor, K_{zt} ; see Section 26.8 and Fig. 26.8-1.
 - Ground elevation factor, K_e ; see Section 26.9 and Table 26.9-1.
 - Enclosure classification; see Section 26.12.
 - Internal pressure coefficient, (GC_{pi}) ; see Section 26.13 and Table 26.13-1.
- Step 4:** Determine velocity pressure exposure coefficient, K_z or K_h ; see Table 26.10-1.
- Step 5:** Determine velocity pressure, q_z or q_h , Eq. (26.10-1).
- Step 6:** Determine external pressure coefficient, (GC_p) , using Fig. 28.3-1 for flat and gable roofs.

User Note: See Commentary Fig. C28.3-2 for guidance on hip roofs.

-
- Step 7:** Calculate wind pressure, p , from Eq. (28.3-1).
-

28.2.1 Wind Load Parameters Specified in Chapter 26. The following wind load parameters shall be determined in accordance with Chapter 26:

- Basic wind speed, V (Section 26.5).
- Wind directionality factor, K_d (Section 26.6).
- Exposure category (Section 26.7).
- Topographic factor, K_{zt} (Section 26.8).
- Ground elevation factor, K_e (Section 26.9).
- Velocity Pressure Exposure Coefficient, K_z or K_h (Section 26.10).
- Enclosure classification (Section 26.12).
- Internal pressure coefficient (GC_{pi}) (Section 26.13).

28.3 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

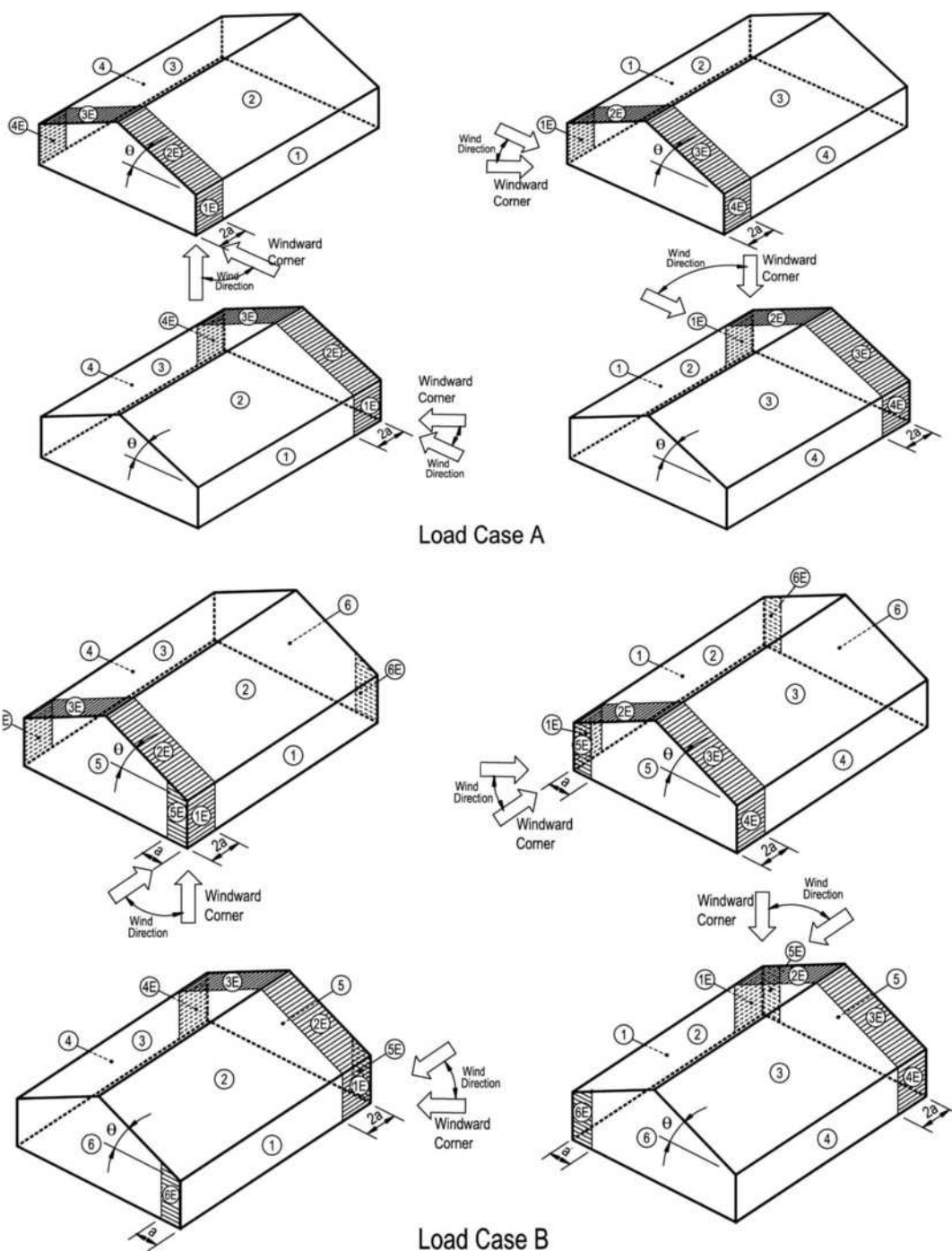
28.3.1 Design Wind Pressure for Low-Rise Buildings. Design wind pressures for the MWFRS of low-rise buildings shall be determined by the following equation:

$$p = q_h[(GC_{pf}) - (GC_{pi})](\text{lb/ft}^2) \quad (28.3-1)$$

$$p = q_h[(GC_{pf}) - (GC_{pi})](\text{N/m}^2) \quad (28.3-1.\text{si})$$

Basic Load Cases

Diagrams



Notation

a 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m).

Exception: For buildings with $\theta = 0$ to 7° and a least horizontal dimension greater than 300 ft (90 m), dimension a shall be limited to a maximum of $0.8h$.

h Mean roof height, in feet (meters), except that eave height shall be used for $\theta \leq 10^\circ$.

θ Angle of plane of roof from horizontal, in degrees.

FIGURE 28.3-1 Main Wind Force Resisting System, Part 1 [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GC_{pf}), for Enclosed and Partially Enclosed Buildings—Low-Rise Walls and Roofs

continues

Load Case A

Roof Angle θ (degrees)	Building Surface							
	1	2	3	4	1E	2E	3E	4E
0–5	0.40	–0.69	–0.37	–0.29	0.61	–1.07	–0.53	–0.43
20	0.53	–0.69	–0.48	–0.43	0.80	–1.07	–0.69	–0.64
30–45	0.56	0.21	–0.43	–0.37	0.69	0.27	–0.53	–0.48
90	0.56	0.56	–0.37	–0.37	0.69	0.69	–0.48	–0.48

Load Case B

Roof Angle θ (degrees)	Building Surface											
	1	2	3	4	5	6	1E	2E	3E	4E	5E	6E
0–90	–0.45	–0.69	–0.37	–0.45	0.40	–0.29	–0.48	–1.07	–0.53	–0.48	0.61	–0.43

Notes

1. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
2. For values of θ other than those shown, linear interpolation is permitted.
3. The building must be designed for all wind directions using the eight loading patterns shown. The load patterns are applied to each building corner in turn as the windward corner.
4. Combinations of external and internal pressures (see Table 26.13-1) shall be evaluated as required to obtain the most severe loadings.
5. For the torsional load cases shown below, the pressures in zones designated with a “T” (1T, 2T, 3T, 4T, 5T, 6T) shall be 25% of the full design wind pressures (zones 1, 2, 3, 4, 5, 6).

EXCEPTION: One-story buildings with h less than or equal to 30 ft (9.1 m), buildings two stories or fewer framed with light-frame construction, and buildings two stories or fewer designed with flexible diaphragms need not be designed for the torsional load cases.

Torsional loading shall apply to all eight basic load patterns using the figures below applied at each windward corner.

6. For purposes of designing a building’s MWFRS, the total horizontal shear shall not be less than that determined by neglecting the wind forces on the roof.

EXCEPTION: This provision does not apply to buildings using moment frames for the MWFRS.

7. For flat roofs, use $\theta = 0^\circ$ and locate the zone 2/3 and zone 2E/3E boundary at the mid-width of the building.
8. The roof pressure coefficient (GC_{pf}), when negative in Zone 2 and 2E, shall be applied in Zone 2/2E for a distance from the edge of roof equal to 0.5 times the horizontal dimension of the building parallel to the direction of the MWFRS being designed or 2.5 times the eave height at the windward wall, whichever is less; the remainder of Zone 2/2E extending to the ridge line shall use the pressure coefficient (GC_{pf}) for Zone 3/3E.

Torsional Load Cases

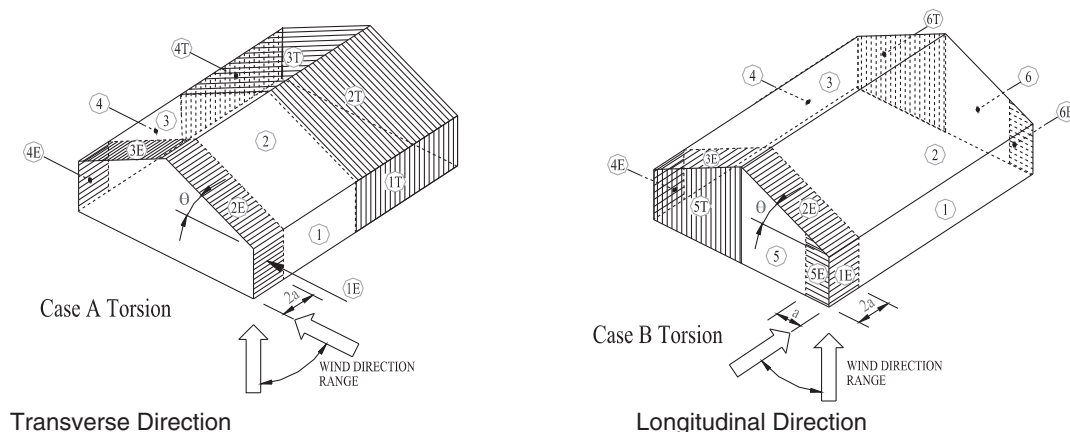
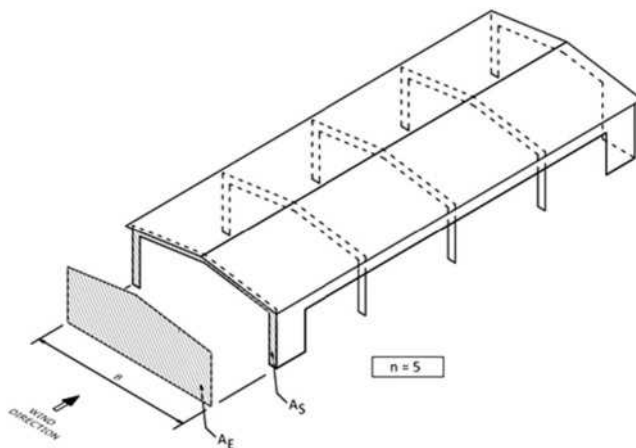


FIGURE 28.3-1 (Continued). Main Wind Force Resisting System, Part 1 [$h \leq 60$ ft ($h \leq 18.3$ m)]: External Pressure Coefficients, (GC_{pf}), for Enclosed and Partially Enclosed Buildings—Low-Rise Walls and Roofs

Diagram



Notation

- B = Width of the building perpendicular to the ridge, in ft (m)
 A_S = Effective solid area of the end wall, i.e., the projected area of any portion of the end wall that would be exposed to the wind
 A_E = Total end wall area for an equivalent enclosed building
 n = Number of frames but shall not be taken as less than $n = 3$

FIGURE 28.3-2 Horizontal Wind Loads on Open or Partially Enclosed Buildings with Transverse Frames and Pitched Roofs: Definitions of Geometric Terminology

where

q_h = velocity pressure evaluated at mean roof height h as defined in Section 26.3;

(GC_{pf}) = external pressure coefficient from Fig. 28.3-1; and

(GC_{pi}) = internal pressure coefficient from Table 26.13-1.

28.3.1.1 External Pressure Coefficients (GC_{pf}). The combined gust-effect factor and external pressure coefficients for low-rise buildings, (GC_{pf}) , are not permitted to be separated.

28.3.2 Parapets. The design wind pressure for the effect of parapets on MWFRS of low-rise buildings with flat, gable, or hip roofs shall be determined by the following equation:

$$p_p = q_p(GC_{pn})(\text{lb/ft}^2) \quad (28.3-2)$$

$$p_p = q_p(GC_{pn})(\text{N/m}^2) \quad (28.3-2.\text{si})$$

where

p_p = combined net pressure on the parapet caused by the combination of the net pressures from the front and back parapet surfaces. Plus (and minus) signs signify net pressure acting toward (and away from) the front (exterior) side of the parapet.

q_p = velocity pressure evaluated at the top of the parapet; and

(GC_{pn}) = combined net pressure coefficient:

= +1.5 for windward parapet; or

= -1.0 for leeward parapet.

28.3.3 Roof Overhangs. The positive external pressure on the bottom surface of windward roof overhangs shall be determined using $GC_p = 0.7$ in combination with the top surface pressures determined using Fig. 28.3-1.

28.3.4 Minimum Design Wind Loads. The wind load to be used in the design of the MWFRS for an enclosed or partially enclosed building shall not be less than 16 lb/ft² (0.77 kN/m²) multiplied by the wall area of the building, and 8 lb/ft²

(0.38 kN/m²) multiplied by the roof area of the building projected onto a vertical plane normal to the assumed wind direction.

28.3.5 Horizontal Wind Loads on Open or Partially Enclosed Buildings with Transverse Frames and Pitched Roofs. A horizontal pressure in the longitudinal direction (parallel to the ridge) that acts in combination with the roof load calculated in Section 27.4.3 for an open or partially enclosed building with transverse frames and a pitched roof ($\theta < 45^\circ$) shall be determined by the following equation:

$$p = q_h[(GC_{pf})_{\text{windward}} - (GC_{pf})_{\text{leeward}}]K_BK_S \quad (28.3-3)$$

where

q_h = velocity pressure evaluated at mean roof height h using the exposure as defined in Section 26.7.3.

(GC_{pf}) = external pressure coefficient given in Fig. 28.3-1 for Load Case B where building surfaces 5 and 5E shall be used to compute the average windward end wall pressure and building surfaces 6 and 6E shall be used to compute the average leeward end wall pressure.

K_B = frame width factor = $1.8 - 0.01B$, $B < 100$ ft ($B < 30.5$ m) or 0.8, $B \geq 100$ ft ($B \geq 30.5$ m).

K_S = shielding factor = $0.60 + 0.073(n - 3) + (1.25 \phi^{1.8})$.

ϕ = solidity ratio = A_S/A_E .

B = width of the building perpendicular to the ridge, in ft (m).

n = number of frames but shall not be taken as less than $n = 3$.

A_S = effective solid area of the end wall, i.e., the projected area of any portion of the end wall that would be exposed to the wind (Fig. 28.3-2).

A_E = total end wall area for an equivalent enclosed building (Fig. 28.3-2).

The total longitudinal force F to be resisted by the MWFRS shall be determined by the following equation:

$$F = pA_E \quad (28.3-4)$$

Eq. (28.3-3) is applicable to buildings with open end walls and with end walls fully or partially enclosed with cladding. For all cases, A_E shall be the area that is equivalent to the end wall fully enclosed. The longitudinal force, F , given by Eq. (28.3-4), represents the total force for which the MWFRS longitudinal bracing shall be designed. The distribution to each sidewall shall be based on force F applied at the centroid of the end wall area A_E .

Fascia load need not be considered separately if fascia areas are included in the A_S calculation.

PART 2: ENCLOSED SIMPLE DIAPHRAGM LOW-RISE BUILDINGS

User Note: Part 2 of Chapter 28 is a simplified method to determine the wind pressure on the MWFRS of enclosed simple diaphragm *low-rise buildings* that have a flat, gable, or hip roof. The wind pressures are *obtained directly from a table* and are applied on horizontal and vertical projected surfaces of the building. This method is a simplification of the Envelope Procedure contained in Part 1 of Chapter 28.

28.4 GENERAL REQUIREMENTS

The steps required for the determination of MWFRS wind loads on enclosed simple diaphragm buildings are shown in Table 28.4-1.

28.4.1 Wind Load Parameters Specified in Chapter 26. The following wind load parameters are specified in Chapter 26:

- Basic wind speed, V (Section 26.5);
- Exposure category (Section 26.7);
- Topographic factor, K_{zt} (Section 26.8)
- Enclosure classification (Section 26.12).

28.5 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

28.5.1 Scope. A building, the design wind loads of which are determined in accordance with this section, shall meet all the conditions of Section 28.5.2. If a building does not meet all of the conditions of Section 28.5.2, then its MWFRS wind loads shall be determined by Part 1 of this chapter, by the Directional Procedure of Chapter 27, or by the Wind Tunnel Procedure of Chapter 31.

28.5.2 Conditions. For the design of MWFRS, the building shall comply with all of the following conditions:

1. The building is a simple diaphragm building as defined in Section 26.2.
2. The building is a low-rise building as defined in Section 26.2.
3. The building is enclosed as defined in Section 26.2 and conforms to the wind-borne debris provisions of Section 26.12.3.
4. The building is a regular-shaped building as defined in Section 26.2.
5. The building is not classified as a flexible building as defined in Section 26.2.

Table 28.4-1 Steps to Determine Wind Loads on MWFRS Simple Diaphragm, Low-Rise Buildings

Step 1: Determine Risk Category of building; see Table 1.5-1.
Step 2: Determine the basic wind speed, V , for applicable Risk Category; see Figs. 26.5-1 and 26.5-2.
Step 3: Determine wind load parameters: • Exposure category B, C, or D; see Section 26.7. • Topographic factor, K_{zt}; see Section 26.8 and Fig. 26.8-1.
Step 4: Enter figure to determine wind pressures for $h = 30$ ft ($h = 9.1$ m), p_{S30} ; see Fig. 28.5-1.
Step 5: Enter figure to determine adjustment for building height and exposure, λ ; see Fig. 28.5-1.
Step 6: Determine adjusted wind pressures, p_s ; see Eq. (28.5-1).

6. The building does not have response characteristics that make it subject to across-wind loading, vortex shedding, instability caused by galloping or flutter; and it does not have a site location for which channeling effects or buffeting in the wake of upwind obstructions warrant special consideration.
7. The building has an approximately symmetrical cross section in each direction with either a flat roof or a gable or hip roof with $\theta \leq 45^\circ$.
8. The building is exempted from torsional load cases as indicated in Note 5 of Fig. 28.3-1, or the torsional load cases defined in Note 5 do not control the design of any of the MWFRS of the building.

28.5.3 Design Wind Loads. Simplified design wind pressures, p_s , for the MWFRS of low-rise, simple diaphragm buildings represent the net pressures (sum of internal and external) to be applied to the horizontal and vertical projections of building surfaces, as shown in Fig. 28.5-1. For the horizontal pressures (Zones A, B, C, D), p_s is the combination of the windward and leeward net pressures. p_s shall be determined by the following equation:

$$p_s = \lambda K_{zt} p_{S30} \quad (28.5-1)$$

where

λ = adjustment factor for building height and exposure from Fig. 28.5-1.

K_{zt} = topographic factor as defined in Section 26.8 evaluated at 0.33 times the mean roof height, $0.33h$.

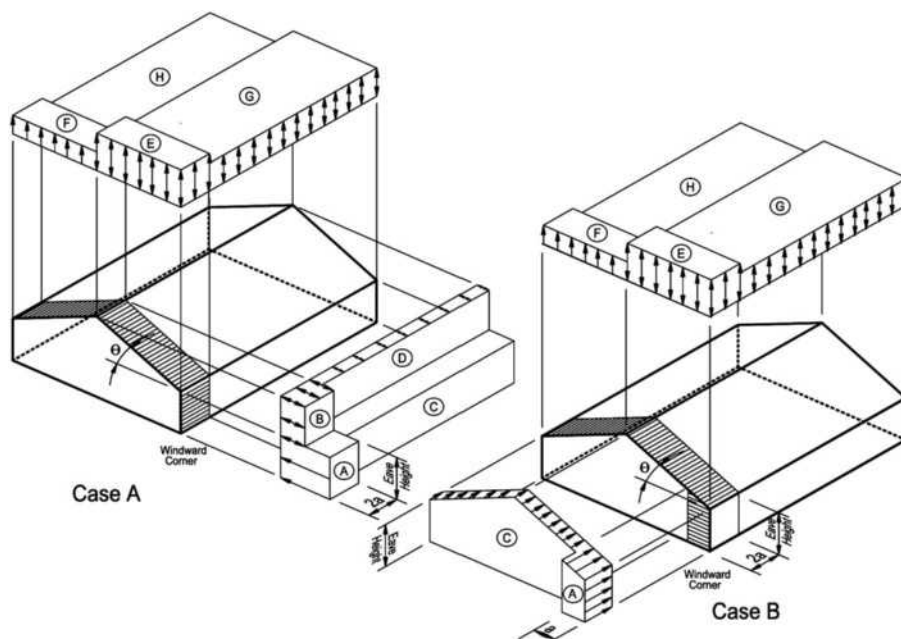
p_{S30} = simplified design wind pressure for Exposure B, at $h = 30$ ft ($h = 9.1$ m) from Fig. 28.5-1.

28.5.4 Minimum Design Wind Loads. The load effects of the design wind pressures from Section 28.5.3 shall not be less than a minimum load defined by assuming the pressures, p_s , for Zones A and C equal to $+16$ lb/ft² (0.77 N/m²), Zones B and D equal to $+8$ lb/ft² (0.38 N/m²), while assuming p_s for Zones E, F, G, and H are equal to 0 lb/ft² (0 N/m²).

28.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

No consensus standards and other documents that shall be considered part of this standard are referenced in this chapter.

Diagrams



Notation

- a 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m).
EXCEPTION: For buildings with $\theta = 0$ to 7° and a least horizontal dimension greater than 300 ft (90 m), dimension a shall be limited to a maximum of $0.8h$.
 h Mean roof height, in ft (m), except that eave height shall be used for roof angles $< 10^\circ$.
 θ Angle of plane of roof from horizontal, in degrees.

Notes

- Pressures shown are applied to the horizontal and vertical projections, for Exposure B, at $h = 30$ ft ($h = 9.1$ m). Adjust to other exposures and heights with adjustment factor λ .
- The load patterns shown shall be applied to each corner of the building in turn as the reference corner (See Fig. 28.3-1).
- For Case B, use $\theta = 0^\circ$.
- Load cases 1 and 2 must be checked for $25^\circ < \theta \leq 45^\circ$. Load case 2 at 25° is provided only for interpolation between 25° and 30° .
- Plus and minus signs signify pressures acting toward and away from the projected surfaces, respectively.
- For roof slopes other than those shown, linear interpolation is permitted.
- The total horizontal load shall not be less than that determined by assuming $p_s = 0$ in Zones B and D.
- Where Zone E or G falls on a roof overhang on the windward side of the building, use E_{OH} and G_{OH} for the pressure on the horizontal projection of the overhang. Overhangs on the leeward and side edges shall have the basic zone pressure applied.
- Unit conversions for tables:

Adjustment Factor for Building Height and Exposure, λ

Mean roof height (ft)	Exposure		
	B	C	D
15	1.00	1.21	1.47
20	1.00	1.29	1.55
25	1.00	1.35	1.61
30	1.00	1.40	1.66
35	1.05	1.45	1.70
40	1.09	1.49	1.74
45	1.12	1.53	1.78
50	1.16	1.56	1.81
55	1.19	1.59	1.84
60	1.22	1.62	1.87

Note: Unit conversions for tables: 1.0 ft = 0.3048 m; 1.0 lb/ft² = 0.0479 kN/m²; 1 mph = 1.6 km/h

FIGURE 28.5-1 Main Wind Force Resisting System, Part 2 [$h \leq 60$ ft ($h \leq 18.3$ m)]: Design Wind Pressures for Enclosed Buildings—Walls and Roofs

continues

Simplified Design Wind Pressure, P_{s30} (psf) for Exposure B at $h = 30$ ft ($h = 9.1$ m)												
Basic Wind Speed (mph)	Roof Angle (degrees)	Load Case	Zones									
			Horizontal Pressures				Vertical Pressures				Overhangs	
			A	B	C	D	E	F	G	H	E_{OH}	G_{OH}
85	0 to 5°	1	11.5	-5.9	7.6	-3.5	-13.8	-7.8	-9.6	-6.1	-19.3	-15.1
	10°	1	12.9	-5.4	8.6	-3.1	-13.8	-8.4	-9.6	-6.5	-19.3	-15.1
	15°	1	14.4	-4.8	9.6	-2.7	-13.8	-9.0	-9.6	-6.9	-19.3	-15.1
	20°	1	15.9	-4.2	10.6	-2.3	-13.8	-9.6	-9.6	-7.3	-19.3	-15.1
	25°	1	14.4	2.3	10.4	2.4	-6.4	-8.7	-4.6	-7.0	-11.9	-10.1
		2	—	—	—	—	-2.4	-4.7	-0.7	-3.0	—	—
	30 to 45	1	12.9	8.8	10.2	7.0	1.0	-7.8	0.3	-6.7	-4.5	-5.2
		2	12.9	8.8	10.2	7.0	5.0	-3.9	4.3	-2.8	-4.5	-5.2
90	0 to 5°	1	12.8	-6.7	8.5	-4.0	-15.4	-8.8	-10.7	-6.8	-21.6	-16.9
	10°	1	14.5	-6.0	9.6	-3.5	-15.4	-9.4	-10.7	-7.2	-21.6	-16.9
	15°	1	16.1	-5.4	10.7	-3.0	-15.4	-10.1	-10.7	-7.7	-21.6	-16.9
	20°	1	17.8	-4.7	11.9	-2.6	-15.4	-10.7	-10.7	-8.1	-21.6	-16.9
	25°	1	16.1	2.6	11.7	2.7	-7.2	-9.8	-5.2	-7.8	-13.3	-11.4
		2	—	—	—	—	-2.7	-5.3	-0.7	-3.4	—	—
	30 to 45	1	14.4	9.9	11.5	7.9	1.1	-8.8	0.4	-7.5	-5.1	-5.8
		2	14.4	9.9	11.5	7.9	5.6	-4.3	4.8	-3.1	-5.1	-5.8
95	0 to 5°	1	14.3	-7.4	9.5	-4.4	-17.2	-9.8	-12.0	-7.6	-24.1	-18.8
	10°	1	16.1	-6.7	10.7	-3.9	-17.2	-10.5	-12.0	-8.1	-24.1	-18.8
	15°	1	18.0	-6.0	12.0	-3.4	-17.2	-11.2	-12.0	-8.6	-24.1	-18.8
	20°	1	19.8	-5.2	13.2	-2.9	-17.2	-12.0	-12.0	-9.1	-24.1	-18.8
	25°	1	18.0	2.9	13.0	3.0	-8.0	-10.9	-5.8	-8.7	-14.9	-12.7
		2	—	—	—	—	-3.0	-5.9	-0.8	-3.8	—	—
	30 to 45	1	16.1	11.0	12.8	8.8	1.2	-9.8	0.4	-8.4	-5.6	-6.5
		2	16.1	11.0	12.8	8.8	6.2	-4.8	5.4	-3.4	-5.6	-6.5
100	0 to 5°	1	15.9	-8.2	10.5	-4.9	-19.1	-10.8	-13.3	-8.4	-26.7	-20.9
	10°	1	17.9	-7.4	11.9	-4.3	-19.1	-11.6	-13.3	-8.9	-26.7	-20.9
	15°	1	19.9	-6.6	13.3	-3.8	-19.1	-12.4	-13.3	-9.5	-26.7	-20.9
	20°	1	22.0	-5.8	14.6	-3.2	-19.1	-13.3	-13.3	-10.1	-26.7	-20.9
	25°	1	19.9	3.2	14.4	3.3	-8.8	-12.0	-6.4	-9.7	-16.5	-14.0
		2	—	—	—	—	-3.4	-6.6	-0.9	-4.2	—	—
	30 to 45	1	17.8	12.2	14.2	9.8	1.4	-10.8	0.5	-9.3	-6.3	-7.2
		2	17.8	12.2	14.2	9.8	6.9	-5.3	5.9	-3.8	-6.3	-7.2

FIGURE 28.5-1 (Continued). Main Wind Force Resisting System, Part 2 [$h \leq 60$ ft ($h \leq 18.3$ m)]: Design Wind Pressures for Enclosed Buildings—Walls and Roofs

continues

Simplified Design Wind Pressure, P_{s30} (psf) for Exposure B at $h = 30$ ft ($h = 9.1$ m)												
Basic Wind Speed (mph)	Roof Angle (degrees)	Load Case	Zones									
			Horizontal Pressures				Vertical Pressures				Overhangs	
			A	B	C	D	E	F	G	H	E_{OH}	G_{OH}
105	0 to 5°	1	17.5	-9.1	11.6	-5.4	-21.0	-11.9	-14.6	-9.2	-29.4	-23.0
	10°	1	19.7	-8.2	13.1	-4.8	-21.0	-12.8	-14.6	-9.9	-29.4	-23.0
	15°	1	22.0	-7.3	14.6	-4.1	-21.0	-13.7	-14.6	-10.5	-29.4	-23.0
	20°	1	24.2	-6.4	16.1	-3.5	-21.0	-14.6	-14.6	-11.1	-29.4	-23.0
	25°	1	21.9	3.5	15.9	3.6	-9.7	-13.3	-7.1	-10.7	-18.2	-15.5
		2	—	—	—	—	-3.7	-7.2	-1.0	-4.6	—	—
	30 to 45	1	19.7	13.4	15.6	10.8	1.5	-11.9	0.5	-10.3	-6.9	-7.9
		2	19.7	13.4	15.6	10.8	7.6	-5.9	6.6	-4.2	-6.9	-7.9
110	0 to 5°	1	19.2	-10.0	12.7	-5.9	-23.1	-13.1	-16.0	-10.1	-32.3	-25.3
	10°	1	21.6	-9.0	14.4	-5.2	-23.1	-14.1	-16.0	-10.8	-32.3	-25.3
	15°	1	24.1	-8.0	16.0	-4.6	-23.1	-15.1	-16.0	-11.5	-32.3	-25.3
	20°	1	26.6	-7.0	17.7	-3.9	-23.1	-16.0	-16.0	-12.2	-32.3	-25.3
	25°	1	24.1	3.9	17.4	4.0	-10.7	-14.6	-7.7	-11.7	-19.9	-17.0
		2	—	—	—	—	-4.1	-7.9	-1.1	-5.1	—	—
	30 to 45	1	21.6	14.8	17.2	11.8	1.7	-13.1	0.6	-11.3	-7.6	-8.7
		2	21.6	14.8	17.2	11.8	8.3	-6.5	7.2	-4.6	-7.6	-8.7
115	0 to 5°	1	21.0	-10.9	13.9	-6.5	-25.2	-14.3	-17.5	-11.1	-35.3	-27.6
	10°	1	23.7	-9.8	15.7	-5.7	-25.2	-15.4	-17.5	-11.8	-35.3	-27.6
	15°	1	26.3	-8.7	17.5	-5.0	-25.2	-16.5	-17.5	-12.6	-35.3	-27.6
	20°	1	29.0	-7.7	19.4	-4.2	-25.2	-17.5	-17.5	-13.3	-35.3	-27.6
	25°	1	26.3	4.2	19.1	4.3	-11.7	-15.9	-8.5	-12.8	-21.8	-18.5
		2	—	—	—	—	-4.4	-8.7	-1.2	-5.5	—	—
	30 to 45	1	23.6	16.1	18.8	12.9	1.8	-14.3	0.6	-12.3	-8.3	-9.5
		2	23.6	16.1	18.8	12.9	9.1	-7.1	7.9	-5.0	-8.3	-9.5
120	0 to 5°	1	22.8	-11.9	15.1	-7.0	-27.4	-15.6	-19.1	-12.1	-38.4	-30.1
	10°	1	25.8	-10.7	17.1	-6.2	-27.4	-16.8	-19.1	-12.9	-38.4	-30.1
	15°	1	28.7	-9.5	19.1	-5.4	-27.4	-17.9	-19.1	-13.7	-38.4	-30.1
	20°	1	31.6	-8.3	21.1	-4.6	-27.4	-19.1	-19.1	-14.5	-38.4	-30.1
	25°	1	28.6	4.6	20.7	4.7	-12.7	-17.3	-9.2	-13.9	-23.7	-20.2
		2	—	—	—	—	-4.8	-9.4	-1.3	-6.0	—	—
	30 to 45	1	25.7	17.6	20.4	14.0	2.0	-15.6	0.7	-13.4	-9.0	-10.3
		2	25.7	17.6	20.4	14.0	9.9	-7.7	8.6	-5.5	-9.0	-10.3
125	0 to 5°	1	24.8	-12.9	16.4	-7.6	-29.8	-16.9	-20.7	-13.1	-41.7	-32.6
	10°	1	27.9	-11.6	18.6	-6.7	-29.8	-18.2	-20.7	-14.0	-41.7	-32.6
	15°	1	31.1	-10.3	20.7	-5.9	-29.8	-19.5	-20.7	-14.8	-41.7	-32.6
	20°	1	34.3	-9.1	22.9	-5.0	-29.8	-20.7	-20.7	-15.7	-41.7	-32.6
	25°	1	31.1	5.0	22.5	5.1	-13.8	-18.8	-10.0	-15.1	-25.7	-21.9
		2	—	—	—	—	-5.2	-10.2	-1.4	-6.6	—	—
	30 to 45	1	27.9	19.1	22.2	15.2	2.1	-16.9	0.7	-14.5	-9.8	-11.2
		2	27.9	19.1	22.2	15.2	10.7	-8.3	9.3	-6.0	-9.8	-11.2

FIGURE 28.5-1 (Continued). Main Wind Force Resisting System, Part 2 [$h \leq 60$ ft ($h \leq 18.3$ m)]: Design Wind Pressures for Enclosed Buildings—Walls and Roofs

continues

Simplified Design Wind Pressure, P_{s30} (psf) for Exposure B at $h = 30$ ft ($h = 9.1$ m)

Basic Wind Speed (mph)	Roof Angle (degrees)	Load Case	Zones									
			Horizontal Pressures				Vertical Pressures				Overhangs	
			A	B	C	D	E	F	G	H	E_{OH}	G_{OH}
130	0 to 5°	1	26.8	-13.9	17.8	-8.2	-32.2	-18.3	-22.4	-14.2	-45.1	-35.3
	10°	1	30.2	-12.5	20.1	-7.3	-32.2	-19.7	-22.4	-15.1	-45.1	-35.3
	15°	1	33.7	-11.2	22.4	-6.4	-32.2	-21.0	-22.4	-16.1	-45.1	-35.3
	20°	1	37.1	-9.8	24.7	-5.4	-32.2	-22.4	-22.4	-17.0	-45.1	-35.3
	25°	1	33.6	5.4	24.3	5.5	-14.9	-20.4	-10.8	-16.4	-27.8	-23.7
		2	—	—	—	—	-5.7	-11.1	-1.5	-7.1	—	—
	30 to 45	1	30.1	20.6	24.0	16.5	2.3	-18.3	0.8	-15.7	-10.6	-12.1
		2	30.1	20.6	24.0	16.5	11.6	-9.0	10.0	-6.4	-10.6	-12.1
140	0 to 5°	1	31.1	-16.1	20.6	-9.6	-37.3	-21.2	-26.0	-16.4	-52.3	-40.9
	10°	1	35.1	-14.5	23.3	-8.5	-37.3	-22.8	-26.0	-17.5	-52.3	-40.9
	15°	1	39.0	-12.9	26.0	-7.4	-37.3	-24.4	-26.0	-18.6	-52.3	-40.9
	20°	1	43.0	-11.4	28.7	-6.3	-37.3	-26.0	-26.0	-19.7	-52.3	-40.9
	25°	1	39.0	6.3	28.2	6.4	-17.3	-23.6	-12.5	-19.0	-32.3	-27.5
		2	—	—	—	—	-6.6	-12.8	-1.8	-8.2	—	—
	30 to 45	1	35.0	23.9	27.8	19.1	2.7	-21.2	0.9	-18.2	-12.3	-14.0
		2	35.0	23.9	27.8	19.1	13.4	-10.5	11.7	-7.5	-12.3	-14.0
150	0 to 5°	1	35.7	-18.5	23.7	-11.0	-42.9	-24.4	-29.8	-18.9	-60.0	-47.0
	10°	1	40.2	-16.7	26.8	-9.7	-42.9	-26.2	-29.8	-20.1	-60.0	-47.0
	15°	1	44.8	-14.9	29.8	-8.5	-42.9	-28.0	-29.8	-21.4	-60.0	-47.0
	20°	1	49.4	-13.0	32.9	-7.2	-42.9	-29.8	-29.8	-22.6	-60.0	-47.0
	25°	1	44.8	7.2	32.4	7.4	-19.9	-27.1	-14.4	-21.8	-37.0	-31.6
		2	—	—	—	—	-7.5	-14.7	-2.1	-9.4	—	—
	30 to 45	1	40.1	27.4	31.9	22.0	3.1	-24.4	1.0	-20.9	-14.1	-16.1
		2	40.1	27.4	31.9	22.0	15.4	-12.0	13.4	-8.6	-14.1	-16.1
160	0 to 5°	1	40.6	-21.1	26.9	-12.5	-48.8	-27.7	-34.0	-21.5	-68.3	-53.5
	10°	1	45.8	-19.0	30.4	-11.1	-48.8	-29.8	-34.0	-22.9	-68.3	-53.5
	15°	1	51.0	-16.9	34.0	-9.6	-48.8	-31.9	-34.0	-24.3	-68.3	-53.5
	20°	1	56.2	-14.8	37.5	-8.2	-48.8	-34.0	-34.0	-25.8	-68.3	-53.5
	25°	1	50.9	8.2	36.9	8.4	-22.6	-30.8	-16.4	-24.8	-42.1	-35.9
		2*	—	—	—	—	-8.6	-16.8	-2.3	-10.7	—	—
	30 to 45	1	45.7	31.2	36.3	25.0	3.5	-27.7	1.2	-23.8	-16.0	-18.3
		2*	45.7	31.2	36.3	25.0	17.6	-13.7	15.2	-9.8	-16.0	-18.3
170	0 to 5°	1	45.8	-23.8	30.4	-14.1	-55.1	-31.3	-38.3	-24.2	-77.1	-60.4
	10°	1	51.7	-21.4	34.4	-12.5	-55.1	-33.6	-38.3	-25.8	-77.1	-60.4
	15°	1	57.6	-19.1	38.3	-10.9	-55.1	-36.0	-38.3	-27.5	-77.1	-60.4
	20°	1	63.4	-16.7	42.3	-9.3	-55.1	-38.3	-38.3	-29.1	-77.1	-60.4
	25°	1	57.5	9.3	41.6	9.5	-25.6	-34.8	-18.5	-28.0	-47.6	-40.5
		2*	—	—	—	—	-9.7	-18.9	-2.6	-12.1	—	—
	30 to 45	1	51.5	35.2	41.0	28.2	4.0	-31.3	1.3	-26.9	-18.1	-20.7
		2*	51.5	35.2	41.0	28.2	19.8	-15.4	17.2	-11.0	-18.1	-20.7

FIGURE 28.5-1 (Continued). Main Wind Force Resisting System, Part 2 [$h \leq 60$ ft ($h \leq 18.3$ m)]: Design Wind Pressures for Enclosed Buildings—Walls and Roofs

continues

Simplified Design Wind Pressure, P_{s30} (psf) for Exposure B at $h = 30$ ft ($h = 9.1$ m)

Basic Wind Speed (mph)	Roof Angle (degrees)	Load Case	Zones									
			Horizontal Pressures				Vertical Pressures				Overhangs	
			A	B	C	D	E	F	G	H	E_{OH}	G_{OH}
180	0 to 5°	1	51.4	-26.7	34.1	-15.8	-61.7	-35.1	-43.0	-27.2	-86.4	-67.7
	10°	1	58.0	-24.0	38.5	-14.0	-61.7	-37.7	-43.0	-29.0	-86.4	-67.7
	15°	1	64.5	-21.4	43.0	-12.2	-61.7	-40.3	-43.0	-30.8	-86.4	-67.7
	20°	1	71.1	-18.8	47.4	-10.4	-61.7	-43.0	-43.0	-32.6	-86.4	-67.7
	25°	1	64.5	10.4	46.7	10.6	-28.6	-39.0	-20.7	-31.4	-53.3	-45.4
		2*	—	—	—	—	-10.9	-21.2	-3.0	-13.6	—	—
	30 to 45	1	57.8	39.5	45.9	31.6	4.4	-35.1	1.5	-30.1	-20.3	-23.2
		2*	57.8	39.5	45.9	31.6	22.2	-17.3	19.3	-12.3	-20.3	-23.2
190	0 to 5°	1	57.2	-29.7	38.0	-17.6	-68.8	-39.1	-47.9	-30.3	-96.3	-75.4
	10°	1	64.6	-26.8	42.9	-15.6	-68.8	-42.0	-47.9	-32.3	-96.3	-75.4
	15°	1	71.9	-23.8	47.9	-13.6	-68.8	-44.9	-47.9	-34.3	-96.3	-75.4
	20°	1	79.2	-20.9	52.8	-11.6	-68.8	-47.9	-47.9	-36.3	-96.3	-75.4
	25°	1	71.8	11.6	52.0	11.8	-31.9	-43.5	-23.1	-34.9	-59.4	-50.6
		2*	—	—	—	—	-12.1	-23.7	-3.3	-15.1	—	—
	30 to 45	1	64.4	44.0	51.2	35.2	5.0	-39.1	1.7	-33.6	-22.6	-25.9
		2*	64.4	44.0	51.2	35.2	24.8	-19.3	21.5	-13.8	-22.6	-25.9
200	0 to 5°	1	63.4	-32.9	42.1	-19.5	-76.2	-43.3	-53.1	-33.5	-106.7	-83.5
	10°	1	71.5	-29.7	47.6	-17.3	-76.2	-46.5	-53.1	-35.8	-106.7	-83.5
	15°	1	79.7	-26.4	53.1	-15.0	-76.2	-49.8	-53.1	-38.0	-106.7	-83.5
	20°	1	87.8	-23.2	58.5	-12.8	-76.2	-53.1	-53.1	-40.2	-106.7	-83.5
	25°	1	79.6	12.8	57.6	13.1	-35.4	-48.2	-25.6	-38.7	-65.9	-56.1
		2*	—	—	—	—	-13.4	-26.2	-3.7	-16.8	—	—
	30 to 45	1	71.3	48.8	56.7	39.0	5.5	-43.3	1.8	-37.2	-25.0	-28.7
		2*	71.3	48.8	56.7	39.0	27.4	-21.3	23.8	-15.2	-25.0	-28.7

* See Note 4.

FIGURE 28.5-1 (Continued). Main Wind Force Resisting System, Part 2 [$h \leq 60$ ft ($h \leq 18.3$ m)]: Design Wind Pressures for Enclosed Buildings—Walls and Roofs